

MOHAMMAD RAHMANITALAB

Department of Mechanical Engineering, Sharif University of technology, Azadi Ave, Tehran, Iran Email: mohammad.rahmanitalab@sharif.edu
Phone: (+98) 9390053773

EDUCATION



PRESENT

B.SC. IN MECHANICAL ENGINEERING, SHARIF UNIVERSITY OF TECHNOLOGY

- GPA: 3.93/4.0 (18.85/20)
- Minor in COMPUTER SCIENCE
- Expected Graduation: June 2025

JUNE 2021

DIPLOMA MATHEMATICS & PHYSICS

- Final Grade: 4.0/4.0 (19.88/20)

RESEARCH INTERESTS

- **Robotics:** Exoskeleton, Surgery Robots, Humanoid Robots, Manipulation
- **Biomechanics:** Neuromuscular, Musculoskeletal
- **Machine Learning:** Reinforcement Learning, Neural Networks, Deep Learning

SKILLS

Languages

- English (Excellent)
- Farsi (Native)

Programming/ Simulation/ CAD

- C, C++ (Intermediate)
- Python (Intermediate)
- MATLAB (Advanced)
- SolidWorks (Advanced)
- MSC ADAMS (Intermediate)
- Twincat3 (Intermediate)

PROJECTS

FALL 2024

Bachelor Project: Motion Analysis on DMD Children

Supervisor: Prof. Saeed Behzadipour

Conducted motion analysis using IMU sensors to develop therapeutic protocols for DMD children, focusing on data collection, analysis, and protocol design (MATLAB).

Summer 2024

Internship Project: Surgical Robot (SINA Robot)

Supervisor: Omid Mahdi Zadeh, Mostafa Ghanbari Kouchaksaraie

Surgical Robotics with an emphasis on kinematics.

Developed analytical solutions for master-slave robotic systems, enhancing precision and responsiveness in surgical applications (MATLAB, MSC ADAMS, Twincat3).

SPRING 2024

Course Project: Automatic Control

Control of a Rotating Governor

Course Project: Dynamics of Machine

Supervisor: Prof. Saeed Behzadipour

Simulation of a mechanism in Simscape

FALL 2023

Course Project: Machine Design II

Supervisor: Prof. Amir Nourani

Design of Gearbox (Gears, Bearings, Pully, Chain)

SPRING 2023

Course Project: Machine Design I

Supervisor: Prof. Amir Nourani

Shafts, Screw, Weldment Design

SPRING 2022

Course Project: Fundamentals of Programming (C)

Supervisor: Prof. Golmohammadi

Twitter App Programming in C

Experience

- ***Sina Robotics and Medical Innovators Co.***
 - ***Full Time: Jun2024-Sep2024***
 - Assisted in the design and testing of robotic components, focusing on improving responsiveness in surgical robotic systems and haptic feedback.

HONOR & AWARDS

- ***Ranked 187th*** in the National University entrance exam among more than 120000 participants.
- ***Ranked 11th*** in the Mechanical Engineering department among 116 participants in class of 2021.

SELECTED COURSES

- | | |
|--|--|
| • <i>Fundamentals of Programming (C): 20/20</i> | • <i>Machine Design II: 17 /20</i> |
| • <i>Statics: 20/20</i> | • <i>Dynamic of Machine: 20 /20</i> |
| • <i>Discrete Math: 20/20</i> | • <i>Automatic Control: 20 /20</i> |
| • <i>Engineering Mathematics: 20/20</i> | • <i>Measurement and Control Systems: 19.8 /20</i> |
| • <i>Dynamics: 19.2/20</i> | • <i>Fundamental Electrical Engineering I: 18.9 /20</i> |
| • <i>Vibrations: 19/20</i> | • <i>Fundamental Electrical Engineering II: 20 /20</i> |
| • <i>Machine Design I: 18.4 /20</i> | • <i>Training 1 (Internship): P_EX</i> |

EXTRA-CURRICULAR ACTIVITIES

ICERS 2024: Contributed to event organization and media coordination as part of the IT & Media Team.
CARE Student Society: Managing media content and team operations to promote student activities.
Mechanical Engineering Department: Website Developer